

1. DISCOUNTED CASH FLOW (DCF) & WACC ARCHITECTURE		INTRINSIC VALUATION & COST OF CAPITAL
Cash Flow Builds (FCFF & FCFE) $FCFF = EBIT \times (1-t) + D\&A - CapEx - \Delta NWC$ $FCFE = Net\ Income + D\&A - CapEx - \Delta NWC + Net\ Debt\ Issued$ FCFF: Cash to all capital providers; discounted via WACC to yield EV FCFE: Levered cash to equity; discounted via K_E to yield Equity Value directly ΔNWC: Excludes cash & short-term debt; growth drains operational cash	Cost of Capital & Beta Unlevering $WACC = (E/V \times K_E) + (D/V \times K_D \times (1-t))$ $CAPM: K_E = R_f + \beta_L \times ERP \mid \beta_U = \beta_L / (1 + (D/E))$ ERP: Equity Risk Premium = Expected Market Return minus Risk-Free Rate Tax Shield: $(1-t)$ captures corporate tax deductibility of interest expense Relever Beta: $\beta_L = \beta_U \times (1 + (1-t)(D/E))$ to match target capital structure	Terminal Value & EV Bridge $Gordon\ Growth\ TV = [FCFF_n \times (1+g)] / (WACC - g)$ $Exit\ Multiple\ TV = Terminal\ EBITDA \times Exit\ Multiple$ Growth Cap: g must not exceed long-term GDP expansion rate (~2–3%) Enterprise Value: $EV = PV(Explicit\ Forecast\ FCFF) + PV(Terminal\ Value)$ Equity Value: $EV - Total\ Debt - Preferred\ Stock - NCI + Cash$
Use cases: Greenfield intrinsic valuation, fairness opinions, establishing baseline floor acquisition bids.		

2. RELATIVE VALUATION & BENCHMARKING SPECTRUM			TRADING COMPS, PRECEDENTS & FOOTBALL FIELD
Trading Comps (Public Multiples) • EV/EBITDA: Capital-structure neutral; primary operating cash proxy • EV/EBIT: Reflects real CapEx and asset depreciation replacement burden • P/E: Distorted by leverage, non-operating items, and tax variance • PEG Ratio: P/E divided by annual EPS growth rate; <1.0 indicates undervaluation • EV/Sales: Benchmarks pre-profit, high-growth expansion companies	Precedent Transactions (Deal Multiples) • Reflects control premium (typically 20%-40% over unaffected share price) • Includes realized strategic buyer synergies and auction cycle competition • Prone to timing distortion (deals closed at cycle peaks carry inflated multiples) • Provides empirical validation benchmarks for board change-of-control approvals	Football Field Valuation Hierarchy • Floor: Liquidation Value < Sum-of-the-Parts (SOTP) / Hard Net Asset Value • Core Base: Public Trading Comps Range < 52-Week High / Low Price Bounds • Intrinsic Target: DCF Valuation Envelope (WACC + Terminal Growth matrix) • Ceiling: Precedent Transactions + Strategic Synergy Full Realization Range	
Use cases: IPO price band discovery, sell-side M&A pitchbooks, target valuation ranges, board fairness evaluations.			

3. M&A MODELING, STRUCTURAL TERMS & PURCHASE ACCOUNTING		PRO-FORMA EPS, TAX STRUCTURING, SYNERGIES & DEAL PROTECTIONS
Accretion / Dilution Mechanics & Consideration Pro-Forma EPS = (Acquirer NI + Target NI + After-Tax Synergies - After-Tax Deal Exp) / Pro-Forma Shares • All-Stock Rule: Accretive if Acquirer P/E > Offer Target P/E; dilutive if Acquirer P/E < Target P/E • Cost of Cash: Foregone After-Tax Cash Yield - Pre-Tax Cash Rate $\times (1 - t)$ • Cost of Debt: After-Tax Interest Rate = $K_d \times (1 - t)$; typically cheaper than equity issuance • Break-even Synergies: Pre-tax savings required to offset EPS dilution = $\text{Dilution Per Share} \times \text{Shares} / (1 - t)$ • Consideration Mix: Cash increases leverage/financial risk; Stock preserves cash but dilutes governance	Synergies, Integration & Transaction Structures • Hard (Cost) Synergies: Headcount rationalization, SG&A overlap, vendor scale discounts, branch consolidation • Soft (Revenue) Synergies: Cross-selling, distribution geographic access, bundled product pricing power • Realization Phase-in: Model on 1–3 year glidepath; haircut revenue gains by 50% for conservative underwriting • Asset Sale vs Stock Purchase: Asset sale provides step-up in tax basis (338(j)(1)(C) election) creating future tax shields • Earnouts & CVRs: Contingent Value Rights bridge bid-ask valuation spreads via future milestone hurdles	Purchase Price Allocation (PPA) & Deal Protections Goodwill = Equity Purchase Price - Net Identifiable Assets (TMV) + Existing Goodwill Write-Off • Asset Step-Up & DTL: Write-up of PP&E and IP increases future D&A; DTL = Asset Step-Up $\times t$ • Break-up / Termination Fees: 2%–4% of deal value paid by target if accepting a superior unsolicited bid • Reverse Break-Up Fees: Paid by buyer if failing antitrust clearance or financing conditions • MAC / MAE Clauses: Material Adverse Change allows buyer walkaway upon unforeseen catastrophic disruption • Collar Structures: Fixed value vs fixed exchange ratio collars hedge pre-closing equity volatility
Use cases: Board change-of-control approvals, negotiating collar structures, underwriting tax step-ups, antitrust clearance modeling.		

4. 3-STATEMENT FINANCIAL MODELING INTERLINKS		STATEMENT DYNAMICS, CASH FLOW MECHANICS & BALANCE SHEET EQUITIES
Income Statement to Cash Flow • Net Income: Bottom-line accrual profit feeds directly as top line of CFO • D&A Addback: Non-cash expenses added back to CFO (tax shield preserved) • Stock-Based Comp: Non-cash expense added back to CFO; impacts diluted shares	Cash Flow to Balance Sheet • Ending Cash: Net Change in Cash (CFO+CFI+CFF) links to Balance Sheet Cash • CapEx: Outflow in CFI links to Gross PP&E (PP&E_end - PP&E_start + CapEx - Depr) • Debt Issuance/Repayment: CFF debt flows link directly to Short/Long-Term Debt	Balance Sheet Retained Earnings Tie Ending RE - Beginning RE + Net Income - Dividends Declared • Working Capital: Operational asset/liability delta in CFO balances BS • Balancing Check: Assets MUST strictly equal Liabilities + Shareholders' Equity
Use cases: LBO debt sweep automation, building dynamic 3-statement integrated models, corporate liquidity stress-testing		

5. LEVERAGED BUYOUT (LBO) MODEL & CREDIT ARCHITECTURE

CAPITAL TRANCHEs, VALUE ATTRIBUTION & DEBT STRUCTURING

Debt Tranches & Pricing Stack	Value Creation Levers & Returns Attribution	Cash Sweep, Covenants & Waterfall Restrictions
Revolving Credit: Working capital liquidity, priced at SOFR + 175-250 bps; undrawn fee Term Loan A (TLA): Amortizing bank debt; 5-yr maturity; SOFR + 200-300 bps (low cost) Term Loan B (TLB): Institutional debt; 7-yr maturity; 1% nominal annual amort; bullet at exit Mezzanine / Sub Debt: Subordinated debt; 10%-14% coupon (Cash + PIK) + equity warrants Sponsor Equity Plug: 30%-45% of Sources - Uses: Common stock + Rollover equity from founder Original Issue Discount (OID): Upfront discount (e.g. 98-99c) reducing net debt proceeds	MoC - Total Exit Equity Proceeds / Initial Sponsor Equity IRR - (MoC^{1/5}) / Holding Period - 1 Rule: 2.0x in 5 yrs ≈ 15% IRR, 2.5x ≈ 20% Deleveraging: Free cash flow swept to retire debt principal, transferring EV to equity EBITDA Growth: Organic volume expansion, pricing, margin expansion, cost takeout Multiple Expansion: Exit EV/EBITDA > Entry multiples; highly sensitive to market cycle Value Attribution: $\Delta \text{Equity} = \Delta \text{Debt Repaid} + \Delta \text{EBITDA} \times \text{Entry Multiple} + \text{Exit Multiple} \Delta \text{Debt} \times \text{Exit EBITDA}$ Dividend Recapitalization: Issue new debt mid-hold to fund sponsor dividend; denisks check	Cash Sweep, Covenants & Waterfall Restrictions Excess Cash Flow (ECF) - Cash Flow Available for Debt Service - Mandatory Amort ECF Sweep: Contractual mandate requiring 50%-100% of residual cash to prepay term debt Maintenance Covenants: Leverage (Total Debt/EBITDA < 5.5x), Interest Coverage (EBITDA/Int > 3.0x) Incurrence Covenants: High-yield debt rules triggered only on events (new debt, M&A, dividends) Cov-Lite Loans: Institutional TLEs lacking maintenance covenants, providing flexibility Restricted Payments Basket: Formula limiting dividends/share buybacks to cumulative retained earnings

Use cases: Underwriting buyout investments, debt capacity stress-testing, management equity pool structuring.

6. VENTURE CAPITAL, CAP TABLES & SAAS METRICS

EARLY-STAGE DILUTION, UNIT ECONOMICS & RUNWAY

Cap Table Mechanics & Dilution	SaaS Unit Economics & Health	Runway & Convertible Instruments
Post-Money Valuation = Pre-Money Valuation + New Capital Investor Ownership % = Investment Amount / Post-Money Valuation Option Pool Shuffles: Pool expanded pre-money, forcing all dilution onto founders Anti-Dilution: Full-ratchet (punitive) vs Broad-Based Weighted Average Liquidation Pref: 1x Non-Participating (standard) vs Participating (double-dip)	SaaS Unit Economics & Health LTV = (ARPU × Gross Margin %) / Churn Rate Benchmarks: LTV : CAC ≥ 3.0x CAC Payback Period < 12 Months Rule of 40: YrY ARR Growth Rate (%) + Free Cash Flow Margin (%) ≥ 40% Net Retention: NRR > 105% signals expansion revenue outpaces logo churn Magic Number: (Quarterly ARR Growth × 4) / Prior Quarter S&M Spend > 0.75x	Runway & Convertible Instruments Runway = Cash Balance / Monthly Net Burn (Target: 18-24 Months) Net Burn: Cash Collections minus Cash Operating Outflows SAFE / Convertible Note: Valuation Cap, Conversion Discount (typically 15%-20%) Down-Round Protections: Pay-to-play provisions, warrant coverage enhancements

Use cases: Series Seed to Series C term sheet negotiations, SaaS cohort retention audits, founder cap table modeling.

7. GLOBAL MARKETS: DERIVATIVES, FOREX & FIXED INCOME

OPTION GREEKS, EXOTICS, FX PARITIES & YIELD CURVE DYNAMICS

Derivatives, Greeks & Volatility Surface	Foreign Exchange (FX), Parity & Carry Trade	Fixed Income, Duration & Term Structure
Put-Call Parity: $C + PV(K) = P + S_0$ Black-Scholes: $C = S_0 \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2)$ Delta (Δ): Hedge ratio ($\partial C / \partial S$); Calls (0 to +1), Puts (-1 to 0); shifts as spot moves Gamma (Γ): Delta convexity ($\partial^2 C / \partial S^2$); peaks at-the-money (ATM); tracks rehedging risk Vega (V): Volatility sensitivity ($\partial C / \partial \sigma$); Theta (Θ): Negative time decay ($\partial C / \partial t$) Rho (ρ): Interest rate sensitivity ($\partial C / \partial r$); higher rates increase call value, reduce put Volatility Smile / Skew: OTM puts trade at implied volatility premium vs OTM calls Interest Rate Swaps: Fixed-for-floating cash settlement against SOFR benchmark curves Credit Default Swaps (CDS): Credit protection contract; Spread ≈ Hazard Rate × (1 - Recovery)	Foreign Exchange (FX), Parity & Carry Trade Covered Interest Parity (CIP): Forward / Spot = $(1 + id) / (1 + if)$ Forward Points: $F - S_0 = S_0 \times [(id - if) \times (\text{Days}/360)] / (1 + if \times (\text{Days}/360))$ Uncovered Interest Parity (UIP): Expected $\% \Delta \text{Spot} \approx id - if$; high yield currencies depreciate Purchasing Power Parity (PPP): Long-term FX moves match inflation deltas: $\% \Delta \text{FX} \approx \pi_{id} - \pi_{if}$ FX Swaps vs Cross-Currency Swaps: FX swaps exchange principal at spot & forward; CCy swaps also exchange periodic floating interest rates across loan lifespans Corporate FX Exposure: Transaction (committed invoices), Translation (foreign subsidiary balance sheet consolidation), Economic/Operating (structural competitive margin risks) FX Carry Trade: Long high-yielding currency funded by low-rate currency; prone to crash risk	Fixed Income, Duration & Term Structure Modified Duration: $D_{mod} = D_{mac} / (1 + YTM/m)$ $\% \Delta \text{Price} \approx -D_{mod} \times \Delta y + 0.5 \times \text{Conv} \times (\Delta y)^2$ DV01 (Dollar Value of a Basis Point): $D_{mod} \times \text{Bond Price} \times 0.0001$ Convexity (Conv): Curvature correction; positive convexity benefits both rate cuts and hikes Yield Curve Shifts: Parallel shift, Steeper (Bull: short yields fall; Bear: long yields spike), Flatterer (Bull: long yields drop; Bear: short rates rise on central bank hikes) Credit Spreads: G-Spread (spread over Treasuries), Z-Spread (zero-volatility spread over spot yield curves), OAS (Option-Adjusted Spread; removes embedded call/put option value) Inverted Curve: 2s10s or 3m10s inversion signals market expectation of impending recession

Use cases: Cross-border M&A currency hedging, managing sponsor debt floating rate risk, structuring high-yield credit tranches.

8. CORE FINANCIAL RATIOS & DUPONT DISSECTION

PROFITABILITY, EFFICIENCY, LEVERAGE & WORKING CAPITAL CYCLES

Profitability	DuPont 3-Step	Leverage & Solvency	Coverage & Liquidity	Operating Cycles
Gross Margin: Gross Profit / Rev EBITDA Margin: EBITDA / Rev Net Margin: Net Income / Rev ROIC: NOPAT / Invested Capital ROCE: EBIT / Capital Employed	ROE = Margin × Turnover × Leverage Margin: Net Income / Sales Turnover: Sales / Assets Leverage: Assets / Equity - Isolates operational vs financial return	Leverage & Solvency Total Leverage: Debt / EBITDA Net Leverage: Net Debt / EBITDA Debt / Equity: Total Debt / Book Eq Asset Coverage: Total Assets / Debt	Coverage & Liquidity Interest Coverage: EBIT / Interest DSCR: NOI / Debt Service Current Ratio: CA / CL Quick Ratio: (Cash + AR) / CL	Operating Cycles DSO: (AR / Rev) × 365 DIO: (Inventory / COGS) × 365 DPO: (AP / COGS) × 365 Cash Conversion Cycle: CCC = DSO + DIO - DPO

Use cases: Operational due diligence, credit rating checks, evaluating covenant headroom, working capital optimization.